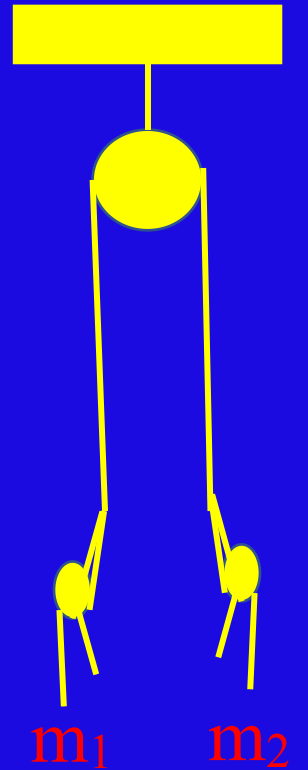


Quiz04: There are two persons with masses m_1 and m_2 ($m_1 > m_2$), respectively. They each hold onto the two ends of the massless ropes hanging over the massless and frictionless pulley. If at the beginning, the distances of both persons away from the pulley are h . Find the distance of the person with the heavier mass away from the pulley when the person with the lighter mass reaches the pulley at the time t after they climb upwards.



Answer:

$$F_T - m_1 g = m_1 a = m_1 \frac{d^2 x_1}{dt^2}$$

$$(m_2 - m_1)g = m_1 \frac{d^2 x_1}{dt^2} - m_2 \frac{d^2 x_2}{dt^2}$$

$$F_T - m_2 g = m_2 a = m_2 \frac{d^2 x_2}{dt^2}$$

$$(m_2 - m_1)gt = m_1 \frac{dx_1}{dt} - m_2 \frac{dx_2}{dt}$$

$$\int_0^t (m_2 - m_1)gt dt = \int_{-h}^{-l} m_1 dx_1 - \int_{-h}^0 m_2 dx_2$$

$$l = (m_1 - m_2)(h + \frac{1}{2}gt^2) / m_1$$

